Continuous Random Variables

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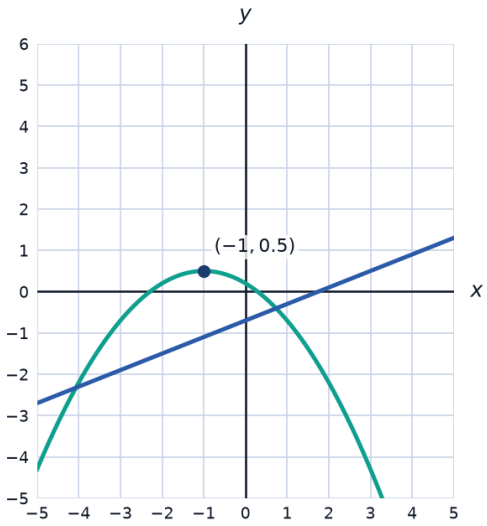
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Verifying a valid probability density function

A continuous random variable X has probability density function $f(x) = \frac{1}{4}$ for $0 \le x \le 4$, and $f(x) = 0$ otherwise. Verify that this is a valid pdf.

2 [Foundational] • Target: Verifying a valid probability density function

Determine the value of c such that $f(x) = c$ for $2 \le x \le 6$, and $f(x) = 0$ otherwise, is a valid pdf.



3 [Foundational] • Target: Finding an unknown constant in a pdf

A continuous random variable X has pdf $f(x) = kx$ for $0 \le x \le 5$, and $f(x) = 0$ otherwise. Find k.

4 [Foundational] • Target: Finding probabilities by integrating the pdf

Using the same pdf, find $P(2 \le X \le 4)$.

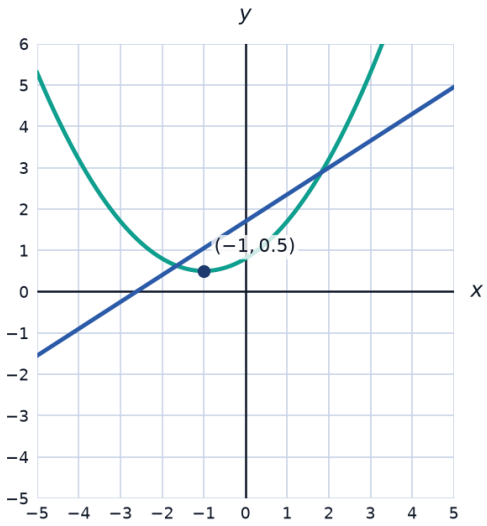
5 [Foundational] • Target: The cumulative distribution function

For the pdf $f(x) = \frac{2}{25}x$ on $0 \le x \le 5$, find the cumulative distribution function $F(x)$.

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6 [Foundational] • Target: The cumulative distribution function

Using $F(x) = \frac{x^2}{25}$, find $P(X \le 3)$ and hence $P(X > 3)$.



7 [Foundational] • Target: Expectation of a continuous random variable

Using $f(x) = \frac{2}{25}x$ for $0 \le x \le 5$, find $E(X)$, using $E(X) = \int x f(x)\,dx$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Expectation of a continuous random variable

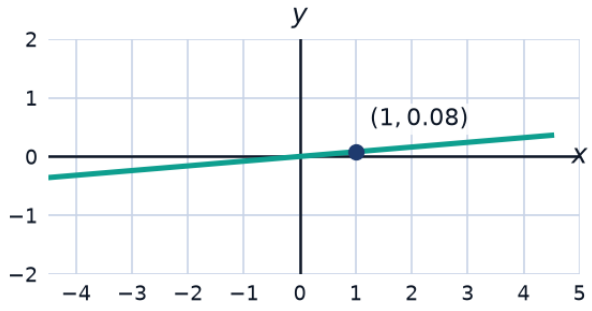
Using the uniform pdf $f(x) = \frac{1}{4}$ for $0 \le x \le 4$, find $E(X)$, and verify it matches the midpoint of the interval.

9 [Intermediate] • Target: Expectation of a continuous random variable

Using $f(x) = \frac{1}{9}x^2$ on $[0, 3]$, find $E(X)$.

10 [Intermediate] • Target: Variance of a continuous random variable

Using $f(x) = \frac{2}{25}x$ on $[0, 5]$ and $E(X) = \frac{10}{3}$ (found earlier), find $\mathrm{Var}(X)$, using $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.



11 [Intermediate] • Target: Variance of a continuous random variable

Using the uniform pdf $f(x) = \frac{1}{4}$ on $[0, 4]$ and $E(X) = 2$ (found earlier), find $\mathrm{Var}(X)$.

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12 [Intermediate] • Target: Variance of a continuous random variable

Using $f(x) = \frac{1}{9}x^2$ on $[0, 3]$ and $E(X) = \frac{9}{4}$ (found earlier), find $\text{Var}(X)$.

13 [Intermediate] • Target: Median and quartiles

Using the cdf $F(x) = \frac{x^2}{25}$ for $0 \le x \le 5$, find the median m of X, solving $F(m) = 0.5$.

14 [Intermediate] • Target: Median and quartiles

Using the same cdf, find the lower quartile Q_1, solving $F(Q_1) = 0.25$.

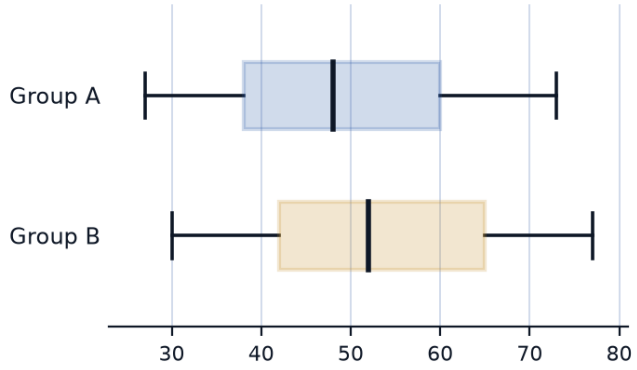
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Median and quartiles • Cross-Lesson: 14.01 Sampling & Estimation

Using $F(x) = \frac{x}{2} - \frac{x^2}{16}$ for $0 \le x \le 4$, find the upper quartile Q_3, solving $F(Q_3) = 0.75$.

16 [Challenging] • Target: Median and quartiles • Cross-Lesson: 15.01 Hypothesis Testing

Using the cdf $F(x) = \frac{x^3}{27}$ for $0 \le x \le 3$, find the median m of X, to three decimal places.



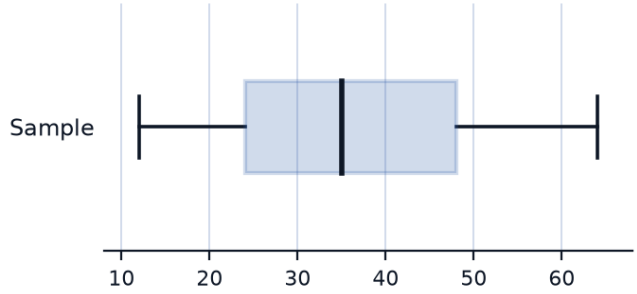
17 [Challenging] • Target: A well-designed word problem: waiting time at a bus stop • Cross-Lesson: 2.01 Exponentials & Logarithms

This question has four parts. The waiting time X (minutes) for a bus is modeled by $f(x) = \frac{3}{64}x^2$ for $0 \le x \le 4$, and $f(x) = 0$ otherwise. (a) Verify this is a valid pdf. (b) Find the cdf $F(x)$, and use it to find the probability a passenger waits more than 3 minutes. (c) Find $E(X)$, the mean waiting time. (d) Find the median waiting time, to three decimal places.

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18 [Challenging] • Target: A well-designed word problem: waiting time at a bus stop • Cross-Lesson: 3.01 Advanced Trigonometry

A continuous random variable X has probability density function $f(x) = kx(2 - x)$ for $0 \le x \le 2$ and $f(x) = 0$ elsewhere. Determine: (a) the constant k; (b) the median of X; (c) the variance $\text{Var}(X)$.



19 [Challenging] • Target: Verifying a valid probability density function • Cross-Lesson: 4.01 Differentiation

A continuous random variable X has probability density function $f(x) = 2e^{-2x}$ for $x \ge 0$ and 0 elsewhere. Find the cumulative distribution function $F(x)$ and determine the exact median of X.

20 [Challenging] • Target: Finding an unknown constant in a pdf • Cross-Lesson: 5.01 Integration

A continuous random variable X has PDF given by $f(x) = \frac{k}{x^2}$ for $1 \le x \le 4$, and 0 elsewhere. (a) Show that $k = \frac{4}{3}$. (b) Find the median of X.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Verifying a valid probability density function

A discrete random variable, X, can take the values 1, 2, 3, 4, and 5. The probability distribution of X is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4, 5\}$, where k is a constant. Find the value of k.

a. $\frac{1}{15}$

b. $\frac{1}{5}$

c. $\frac{1}{3}$

d. 1

22 [Foundational] • Target: Finding an unknown constant in a pdf

A discrete random variable X has probability distribution given by $P(X = 1) = k$, $P(X = 2) = k$, $P(X = 3) = 2k$, where k is a constant. Find k.

a. 1/3

b. 1/4

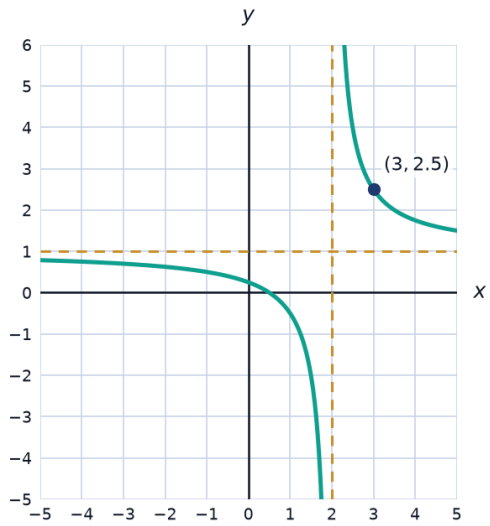
c. 1/5

d. 1/2

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23 [Foundational] • Target: Finding probabilities by integrating the pdf

The rational function f is defined for all real $xotin\{-1, 2\}$ by $f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$ where k is a constant. The partial fraction decomposition of $f(x)$ takes the form $A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$ where $A, B, C,$ and D are real constants. Given that $C = 2$, find the value of k.



a. -3

b. 1

c. 3

d. 24

24 [Foundational] • Target: The cumulative distribution function

A discrete random variable X has distribution $P(X = 1) = 2/5, P(X = 2) = 1/5, P(X = 3) = 2/5$. Given $Y = 5X - 3$, find $E(Y)$.

a. 6

b. 7

c. 8

d. 13

25 [Foundational] • Target: Using the cdf to find probabilities

A discrete random variable X takes values $0, 3, 4$ with probabilities $\frac{1}{10}, \frac{3}{10}, \frac{3}{5}$ respectively. Find $E(X)$.

a. 1

b. 6

c. $\frac{33}{10}$

d. 2

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26 [Foundational] • Target: Expectation of a continuous random variable

A discrete random variable X has distribution $P(X=1)=1/10, P(X=2)=2/5, P(X=3)=1/2$. Find $\text{Var}(X)$.

a. 9/13

b. 11/25

c. 26/37

d. 9/10

27 [Foundational] • Target: Variance of a continuous random variable

A discrete random variable X has distribution $P(X=1)=2/5, P(X=2)=3/10, P(X=3)=3/10$. State $P(X=2)$.

a. 3/4

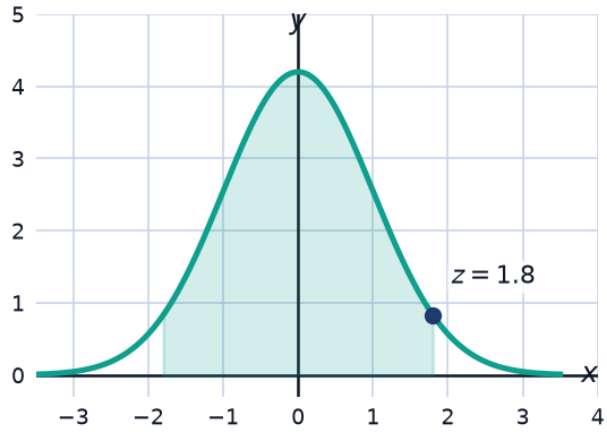
b. 8/9

c. 3/10

d. 11/15

28 [Foundational] • Target: Median and quartiles

A random variable $X \sim B(51, 0.4)$ is approximated by a normal distribution. State the variance of the approximating normal distribution.



a. 12.24

b. 15.24

c. 9.24

d. 19.24

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: waiting time at a bus stop

The rational function $f(x)$ is defined by $f(x) = \frac{5x+7}{(x-1)(x+3)}, \quad xeq1, \ xeq-3.$ Which of the following gives $f(x)$ expressed in partial fractions?

a. $\frac{3}{x-1} + \frac{2}{x+3}$

b. $\frac{2}{x-1} + \frac{3}{x+3}$

c. $\frac{3}{x-1} - \frac{2}{x+3}$

d. $\frac{4}{x-1} + \frac{1}{x+3}$

30 [Intermediate] • Target: Verifying a valid probability density function

Formulate the null and alternative hypotheses for the following scenario: "A company claims its average delivery time is 3 days. A customer believes it takes longer."

a. $H_0 : \mu = 10, \ H_1 : \mu < 10$

b. $H_0 : \mu = 70, \ H_1 : \mu > 70$

c. $H_0 : \mu = 500, \ H_1 : \mu < 500$

d. $H_0 : \mu = 3, \ H_1 : \mu > 3$

31 [Intermediate] • Target: Finding an unknown constant in a pdf

Find the constant term of the quotient when $4x^2 + 6x + 3$ is divided by $(x + 1)$.

a. 2

b. 1

c. 4

d. 3

32 [Intermediate] • Target: Finding probabilities by integrating the pdf

Consider the exponential growth model described by the function $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. 0.347

b. 0.416

c. 0.301

d. 0.477

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33 [Intermediate] • Target: The cumulative distribution function

A random variable $X \sim B(100, 0.6)$ is approximated by a normal distribution. State the mean of the approximating normal distribution.

a. 58.0

b. 65.0

c. 60

d. 62.0

34 [Intermediate] • Target: Using the cdf to find probabilities

A discrete random variable has $P(X = 1) = k$, $P(X = 2) = 2k$, $P(X = 3) = 3k$. Find k.

a. $\frac{5}{12}$

b. $\frac{2}{3}$

c. $\frac{1}{6}$

d. $\frac{5}{6}$

35 [Intermediate] • Target: Expectation of a continuous random variable

A random variable X equals 2 with probability $\frac{1}{5}$ and 4 with probability $\frac{4}{5}$. Find $\mathrm{Var}(X)$.

a. $\frac{2}{5}$

b. $\frac{16}{25}$

c. $\frac{33}{25}$

d. $\frac{39}{50}$

36 [Intermediate] • Target: Variance of a continuous random variable

A continuous function f and a constant a satisfy $\int_a^x f(t)\,dt = e^x - e^5$ for all x, with $a > 0$. Find a.

a. 25

b. -5

c. 10

d. 5

37 [Intermediate] • Target: Median and quartiles

A discrete random variable X is uniformly distributed on $\{1, 2, \ldots, 7\}$. Find $E(X)$.

a. 9/2

b. 3

c. 4

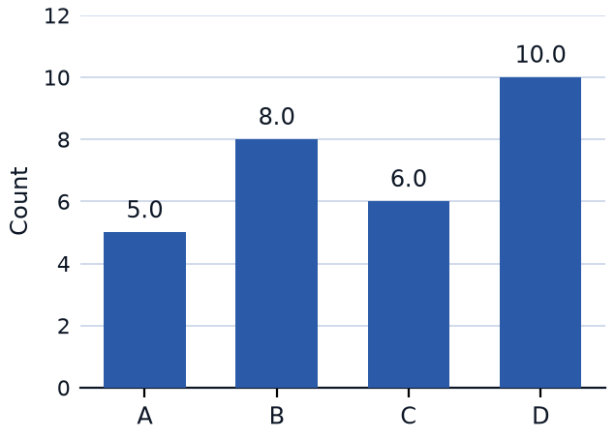
d. 5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A well-designed word problem: waiting time at a bus stop • Cross-Lesson: 3.01 Advanced Trigonometry

The histogram below shows the distribution of heights of a group of students. The total number of students in the group is 50. If the frequency density for the class interval $60 \le h < 70$ is 1.2, find the value of k.



a. 10

b. 12

c. 15

d. 20

39 [Challenging] • Target: Verifying a valid probability density function • Cross-Lesson: 4.01 Differentiation

The complex number z satisfies the equation $|z - (6 + 8\mathrm{i})| = 4$. Given that z^* denotes the complex conjugate of z, determine the maximum possible value of $|z - \mathrm{i}z^*|$.

a. $8 + 2\sqrt{2}$

b. $8 - 2\sqrt{2}$

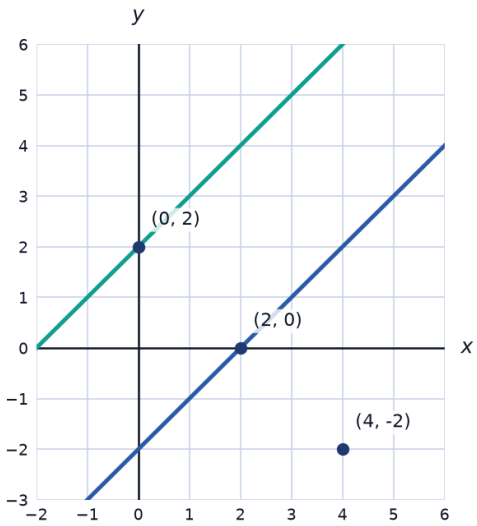
c. $4\sqrt{2} + 2$

d. $4 + \sqrt{2}$

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40 **[Challenging]** • **Target: Finding an unknown constant in a pdf** • **Cross-Lesson: 5.01 Integration**

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$?



a. 17

b. 7

c. 2.4

d. -7

41 **[Challenging]** • **Target: Finding probabilities by integrating the pdf** • **Cross-Lesson: 6.01 Sequences & Series**

Which of the following is the correct expression of $\frac{5x+1}{(x-1)(x+2)}$ in partial fractions?

a. $\frac{2}{x-1} + \frac{3}{x+2}$

b. $\frac{3}{x-1} + \frac{2}{x+2}$

c. $\frac{2}{x-1} - \frac{3}{x+2}$

d. $\frac{1}{x-1} + \frac{4}{x+2}$

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42 [Challenging] • Target: The cumulative distribution function • Cross-Lesson: 7.01 Vectors

Find the exact real solution to the equation $2e^{2x} + 5e^{x} - 12 = 0$.

a. $x = \ln\left(\frac{3}{2}\right)$

b. $x = \ln 4$

c. $x = \frac{3}{2}$

d. $x = \ln\left(\frac{3}{2}\right)$ and $x = \ln 4$

43 [Challenging] • Target: Using the cdf to find probabilities • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

The complex number z satisfies the equation $z(2 - \mathrm{i}) = 7 + 4\mathrm{i}$. Express z in the form $x + \mathrm{i}y$, where x and y are real.

a. $2 + 3\mathrm{i}$

b. $\frac{18}{5} + \frac{1}{5}\mathrm{i}$

c. $2 - 3\mathrm{i}$

d. $\frac{10}{3} + 5\mathrm{i}$

44 [Challenging] • Target: Expectation of a continuous random variable • Cross-Lesson: 9.01 Complex Numbers

Find the exact value of the definite integral $\int_0^{\ln 2} \mathrm{e}^{2x}\,\mathrm{d}x$.

a. $\frac{3}{2}$

b. 3

c. $\frac{1}{2}$

d. $\frac{7}{2}$

45 [Challenging] • Target: Variance of a continuous random variable • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Formulate the null and alternative hypotheses for the following scenario: "A drug is claimed to reduce blood pressure by 10 units on average. A skeptic thinks the true reduction is less than 10."

a. $H_0 : \mu = 250,\ H_1 : \mu e 250$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 10,\ H_1 : \mu < 10$

d. $H_0 : \mu = 500,\ H_1 : \mu < 500$

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